Supplement:
Engineering What an Optical Sensor Cannot See

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July 24, 2026

This Supplement is an audit trail, not an alternate manuscript. It contains the full proofs of the four main-text results (S1–S4), the model and protocol capsule with engines, seeds, and commit hashes (S6), the theorem-consistent honest negatives (S5), and the frozen three-ledger scaling-gate (TLSG) table (S4.3). Every number traces to a single committed frozen artifact recorded in `paper2_optica/CLAIM_SOURCE_MATRIX.md`; no datum was generated for this document. Equation, section, table, and figure labels carry an "S" prefix; bare references point to the main article.

Contents

S1 Statistical Noether wall: proof and periodic table

Let a scene x lie on a smooth parameter manifold $\mathcal{X}$; let the complete observed record have a dominated law P_x with density $p(y \mid x)$; and let a Lie group G act smoothly on $\mathcal{X}$ with infinitesimal generator $X_\xi(x)$ for ξ in the Lie algebra.

Theorem 1 (Statistical Noether wall)**.** *If $P_{g\cdot x} = P_x$ for every $g \in G$, then for every algebra element ξ*

$$D_x \log p(Y \mid x)[X_\xi(x)] = 0 \quad a.s., \qquad I_x\, X_\xi(x) = 0. \tag{S1}$$

Consequently every f-divergence, Chernoff distance, and quotient information jet between two points on a common orbit is zero, and no estimator, moment order, exposure count, or reconstruction prior can distinguish points on that orbit.

Proof. Invariance gives $\log p(y \mid \exp(t\xi)\cdot x) = \log p(y \mid x)$ for all t and almost every y. Differentiating at $t = 0$ yields the directional score identity (S1). Taking the outer product and expectation gives $I_x X_\xi = \mathbb{E}[(D_x \log p\,[X_\xi])(\cdot)] = 0$. Because the complete laws coincide at every finite orbit displacement $P_{g\cdot x} = P_x$, any f-divergence $\mathcal{D}_f(P_{g\cdot x} \| P_x) = \mathcal{D}_f(P_x \| P_x) = 0$; the same holds for the Chernoff distance and for any nuisance-profiled quotient jet, which is an infimum of such divergences. □

Lemma 1 (Local converse)**.** *Let $\mathcal{D}_x$ be a smooth constant-rank distribution that is integrable and satisfies $D_x \log p(y \mid x)[v] = 0$ for every $v \in \mathcal{D}_x$, almost every y, and every x in a connected neighborhood. Then $p(y \mid x)$ is constant along each integral leaf of $\mathcal{D}$; the leaves are local blindness orbits.*

Table S1. Wall periodic table for an optical chain. The first two rows are the walls confirmed here; the mode-unitary and parity rows are immediate consequences of Theorem 1 (Section S1).

Wall class	Invariance / group	Exact null	Breaking operator
Support	additive translations by $\ker A$	scene modes beyond the transfer range	finite window, aliasing, nonlinear moment
Unitary energy	$E \to UE$ under total-energy collection	every lossless mode mixing, incl. phase-only	NA clipping, mode-selective collection
Mode-unitary	full U on spatial modes	any unitary mode transform	mode sorter / analyzer
Global phase	$E \to e^{i\phi}E$ under intensity detection	absolute optical phase	coherent reference
Parity/exchange	$x(r) \to x(-r)$, symmetric chain	odd tangent modes at a parity-even baseline	asymmetric code or pupil
Polarization-unitary	local/global $SU(2)$ under intensity bucket	polarization rotation at fixed intensity	analyzer, birefringent reference

Proof. By Frobenius integrability the distribution has integral leaves. Along any smooth path $\gamma(t)$ inside a leaf, $\frac{d}{dt}\log p(y \mid \gamma(t)) = D_x \log p\,[\dot\gamma] = 0$ for almost every y, so $p(y \mid \cdot)$ is constant on the leaf. Hence, under ordinary smoothness and integrability, continuous local walls are exactly local gauge symmetries of the statistical experiment. □

Remark 1. A symmetry of the underlying wave equation does not by itself create a wall; a wall appears only when the full measurement law inherits the induced parameter action. Discrete degeneracies—parity, conjugation, source–detector exchange—may produce globally indistinguishable pairs without a nonzero infinitesimal tangent; these are walls but not Lie-generated local nulls, and are outside Theorem 1.

The two confirmed walls as group actions. The support wall is the additive group of the unmeasured subspace: for a bandlimited mean operator A, every $h \in \ker A$ gives $x \mapsto x + h$ with $P^{\mathrm{mean}}_{x+h} = P^{\mathrm{mean}}_{x}$; a hard field pupil restores it to a maximum field leak beyond $2kR$ of 6.6×10^{-16} (KT1, `@19338a4`). The energy wall is the unitary orbit of a lossless full-aperture bucket $B(E) = \|E\|_2^2$, with phase-only change the diagonal subgroup; the measured null is 2.18×10^{-16} (IT1, `@f13a323`).

There is no finite universal list independent of apparatus: the complete local list for a declared experiment is the integrable kernel distribution of its score map, and Theorem 1 supplies the classification rule.

S2 Blindness capacity: proof, robustness, and noncomposability

Let $L : \mathcal{C} \to \mathcal{Y}$ be a calibrated linear leak operator with the code norm whitened by the desired in-band metric, and write singular values ascending $\sigma_1 \le \cdots \le \sigma_n$. With $\beta_d(L) = \min_{\dim S = d} \max_{c \in S, \|c\|=1} \|Lc\|$:

Theorem 2 (Blindness capacity). *$\beta_d(L) = \sigma_d(L)$, attained by the span of the bottom d right singular vectors. For a generalized relative-leak certificate with metric $B^\dagger B$, $\beta_d = \sqrt{\lambda_d}$ with $L^\dagger L\, v = \lambda\, B^\dagger B\, v$.*

Proof. Let $v_1, \ldots, v_n$ be right singular vectors with singular values $\sigma_1 \le \cdots \le \sigma_n$ and set $S^\star = \mathrm{span}\{v_1, \ldots, v_d\}$. For $c \in S^\star$ with $\|c\| = 1$, $\|Lc\|^2 = \sum_{i \le d} \sigma_i^2 |\langle c, v_i\rangle|^2 \le \sigma_d^2$, so $\beta_d \le \sigma_d$. Conversely, any d-dimensional S intersects $\mathrm{span}\{v_d, \ldots, v_n\}$ (dimension $n - d + 1$) nontrivially, by a dimension count; a unit vector c in the intersection has $\|Lc\|^2 = \sum_{i \ge d} \sigma_i^2 |\langle c, v_i\rangle|^2 \ge \sigma_d^2$, so $\beta_d \ge \sigma_d$. Hence $\beta_d = \sigma_d$. The generalized statement is the same argument for the pencil $(L^\dagger L, B^\dagger B)$ after the congruence $B^\dagger B = R^\dagger R$. □

Theorem 3 (Robust calibration certificate). *If a calibration returns $\hat L$ with $\|L - \hat L\|_{\mathrm{op}} \le \delta$ and S_d is the bottom-d right singular subspace of $\hat L$, then*

$$\max_{c \in S_d,\ \|c\|=1} \|Lc\| \;\le\; \sigma_d(\hat L) + \delta. \tag{S2}$$

Proof. For $c \in S_d$ with $\|c\| = 1$, $\|Lc\| \le \|\hat{L}c\| + \|(L - \hat{L})c\| \le \sigma_d(\hat{L}) + \delta$, using Theorem 2 for $\hat{L}$ on S_d and the operator-norm bound. □

Equation (S2) converts one calibration run into a metrological cannot-see guarantee: it is the ingredient separating a numerical null space from a certified instrument property. TLSG arm B measures it directly (Section S4.3).

Theorem 4 (Noncomposability)**.** *For leak routes $L_1, \dots, L_k$ with weights w_i, the simultaneous root-sum-square certificate is governed by the stacked operator $L_{\text{stack}} = [w_1 L_1; \dots; w_k L_k]$, so*

$$\beta_d^2 = \lambda_d^{\uparrow}\Big(\textstyle\sum_i w_i^2 L_i^{\dagger} L_i \Big). \tag{S3}$$

The certificate factors are not multiplicative; multiplication occurs only for serial contractions whose singular-vector chains align.

Proof. $L_{\text{stack}}^{\dagger} L_{\text{stack}} = \sum_i w_i^2 L_i^{\dagger} L_i$ by block computation; apply Theorem 2 to L_{stack}. Because the bottom eigenvector of a sum of positive-semidefinite forms is not the product of the individual bottom subspaces, the d-capacities do not multiply. □

Two committed instances of (S3)**.** The two "$\to$ 24" witnesses are distinct and are not conflated. (i) IT5 (`@c8a0487`): the DPSS hardware route does not stack with the SVD code route—the single-plane certified dimension at 10^{-4} drops $80 \to 24$, because the window rotates and reshapes the operator. (ii) IT4b (`@5666d1e`): Tukey apodization is an independent witness of the same law, with joint certified dimension $35 \to 24$ and a $93\times$ worse relative leak at $d = 64$ (4.66×10^{-2} apodized versus 4.99×10^{-4} non-apodized; a rim effect). These are different operators and different splits.

Extra spectra. The full ascending generalized-leak spectrum σ_d versus d (IT4b, `@5666d1e`) runs from $\sigma_1 = 1.43 \times 10^{-6}$ through $\sigma_{64} = 4.99 \times 10^{-4}$ to $\sigma_{120} = 1.13 \times 10^{-1}$; the plotted capacity curve of Fig. 1 (center) is a 30-point sampling of this map. Two capacity points anchor it: single-plane $d@10^{-5} = 80$, and joint (four z_1 planes) $d@10^{-4} = 35$, $d@10^{-5} = 8$, $d@10^{-3} = 80$, so the single-plane requirement buys 45 extra certified dimensions. The $d = 64$ subspace has worst-direction leak 4.99×10^{-4} and operational random-code RMS leak 1.68×10^{-4} (threefold below the worst direction), ridge-stable ($d@10^{-4} = 35$ across regularizers $0/10^{-8}/10^{-6}$). Certified-blind codes keep a second sensing channel: eigenvalue ratio $\lambda_{\text{random}}/\lambda_{\text{blind}} = 1.20$, operational mean leak $219\times$ lower (2.58×10^{-4} versus 5.64×10^{-2}; IT4, `@33b8ef2`).

S3 Jet transmutation: flow equations and specialization

Let the profiled Chernoff distance carry two leading operators,

$$C(\varepsilon, \gamma) = c_1\, \gamma^{2a_1} \varepsilon^{2m_1} + c_2\, \gamma^{2a_2} \varepsilon^{2m_2}, \qquad 0 < m_1 < m_2, \tag{S4}$$

with a geometric gain γ (the contact distance z_2 in the near-contact case), and set $y = (c_2/c_1)\gamma^{2(a_2 - a_1)} \varepsilon^{2(m_2 - m_1)}$. The effective orders are $m_{\text{eff}} = \frac{1}{2}\, \partial_{\log \varepsilon} \log C = (m_1 + m_2 y)/(1+y)$ and $a_{\text{eff}} = \frac{1}{2}\, \partial_{\log \gamma} \log C = (a_1 + a_2 y)/(1+y)$, and a direct differentiation gives the exact flow

$$\frac{\partial m_{\text{eff}}}{\partial \log \varepsilon} = 2(m_{\text{eff}} - m_1)(m_2 - m_{\text{eff}}), \qquad \frac{\partial m_{\text{eff}}}{\partial \log \gamma} = \frac{2(a_2 - a_1)}{m_2 - m_1}(m_{\text{eff}} - m_1)(m_2 - m_{\text{eff}}), \tag{S5}$$

with fixed points m_1 (relevant as $\varepsilon \to 0$) and m_2.

Corollary 1 (Jet transmutation)**.** *A lower-order symmetry-breaking channel with coefficient $c_1 > 0$ drives $m_{\text{eff}} \to m_1$; zeroing c_1 by an exact wall, a certified code, or a nuisance projection places the experiment exactly on the m_2 manifold without changing the scene perturbation.*

For the near-contact optical case $m_1 = 1, a_1 = 0, m_2 = 2, a_2 = 2$ (covariance Chernoff information is $O(z_2^4 \varepsilon^4)$),

$$\frac{\partial m}{\partial \log \varepsilon} = 2(m-1)(2-m), \qquad \frac{\partial m}{\partial \log z_2} = 4(m-1)(2-m), \tag{S6}$$

so that $y \propto \varepsilon^2 z_2^4$, giving $\varepsilon\, z_2^2 = \text{const}$ and $z_{2,*} \propto \varepsilon^{-1/2}$, the geometric form of the KT2 crossover $\varepsilon_{\text{cross}} = \sqrt{c_1/c_2}$. The measured realizations (KT2, `@05f1029`): projected covariance slopes $[4, 4, 4, 4]$; unprojected low-amplitude slopes $[2.136, 2.20, 2.117, 2.176]$; orthogonality residuals in $[1.45, 5.09] \times 10^{-18}$; generic-direction covariance slopes $[1.576, 2.284, 1.576, 2.285]$. The cross-paper integer anchors are the companion KL slopes 2.038 (generic) and 4.000 (first-order-orthogonal) with coefficient agreement $1.07\%/0.00\%$ (JET_TEST, `@1bf29f1`; Ref. [COMPANION]).

S4 Three-ledger certifiability: proof and scaling gate

S4.1 Nuisance-profiled LAN reduction

One independent bank yields $Y_t \sim \mathcal{N}(0, \Sigma_\varepsilon)$, $t = 1, \ldots, T$. Whiten the null covariance and project every admitted nuisance tangent in the Gaussian covariance metric, giving the efficient displacement $A_\perp = \Pi_\perp[\Sigma_0^{-1/2}(\Sigma_\varepsilon - \Sigma_0)\Sigma_0^{-1/2}]$. Let the efficient symmetric tangent $\mathcal{H}$ have dimension p with basis $B_1, \ldots, B_p$ normalized by $\frac{1}{2}\operatorname{tr}(B_j B_k) = \delta_{jk}$, and write $A_\perp = \sum_j \theta_j B_j$, $\lambda = \|\theta\|^2 = \frac{1}{2}\|A_\perp\|_F^2$. The single-bank score $s_j(Y) = \frac{1}{2}[Y^\top B_j Y - \operatorname{tr}(B_j)]$ has $\mathbb{E}_0 s = 0$, $\operatorname{Cov}_0(s) = I_p$, and $\mathbb{E}_A s = \theta + o(\|\theta\|)$, so the normalized aggregate score is locally

$$Z_T = T^{-1/2} \textstyle\sum_t s(Y_t) \;\approx\; \mathcal{N}(\sqrt{T}\,\theta,\, I_p). \tag{S7}$$

S4.2 The three ledgers

Ledger I (known direction). For a known unit direction $u = \theta/\|\theta\|$, the matched score $\nu = u^\top Z_T$ has $\nu \sim \mathcal{N}(0,1)$ under H_0 and $\mathcal{N}(\sqrt{T\lambda}, 1)$ under H_1, so fixed nontrivial power requires and attains $T\lambda = O(1)$.

Ledger II (blind existence). If only $\|\theta\|^2 = \lambda$ is known, the invariant $Q_T = \|Z_T\|^2$ is χ_p^2 under H_0 (mean p, SD $\sqrt{2p}$) and $\chi_p^2(T\lambda)$ under H_1 (mean $p + T\lambda$), so the norm test separates the hypotheses when $T\lambda \gg \sqrt{p}$.

Lemma 2 (Minimax lower bound)**.** *Mixing the alternative uniformly over directions of norm $r = \sqrt{T\lambda}$, the χ^2-divergence of the mixture from the null satisfies $\chi^2 + 1 = \mathbb{E}\exp(\mu^\top \mu')$ for independent mixture directions μ, μ', with $\mathbb{E}(\mu^\top \mu') = 0$ and $\operatorname{Var}(\mu^\top \mu') = r^4/p$. If $r^4/p = T^2\lambda^2/p \to 0$ the divergence tends to zero, hence total variation tends to zero and every test has asymptotic error sum approaching one. No direction-agnostic procedure succeeds below $T\lambda \sim \sqrt{p}$; the norm test attains this scale.*

Ledger III (estimate the direction). The efficient estimator $\hat\theta = Z_T/\sqrt{T}$ has exact local risk $\mathbb{E}\|\hat\theta - \theta\|^2 = p/T$, matching the Cramér–Rao bound, so relative squared error below a constant requires $T\lambda \sim p$. Hence

$$\text{known} : \text{blind} : \text{full} \;=\; 1 : \sqrt{p} : p. \tag{S8}$$

Composing with the physical jet $A_\perp = \kappa\gamma^a\varepsilon^m B$ ($\frac{1}{2}\|B\|_F^2 = 1$; the geometric gain γ is the contact distance z_2 in the near-contact case) gives $\lambda = \kappa^2\gamma^{2a}\varepsilon^{2m}$ and $T_{\text{known}} \asymp \kappa^{-2}\gamma^{-2a}\varepsilon^{-2m}$, $T_{\text{blind}} \asymp \sqrt{p}\,T_{\text{known}}$, $T_{\text{est}} \asymp p\,T_{\text{known}}$; adaptation multiplies the prefactor and leaves the physical exponent $2m$ unchanged. Boundary cases: at $p = 1$ the three ledgers coincide; an exact wall $B = 0$ gives $\lambda = 0$ and no finite T succeeds; a sparse admitted class replaces p by its metric entropy.

S4.3 Frozen scaling gate (TLSG)

The gate is the falsification experiment above plus a calibration-transfer arm for Eq. (S2). Arm A sweeps $M \in \{8, 16, 32, 64\}$ and $p_{\text{eff}} \in \{10, 36, 136, 528\}$ (eight (M, p) cells), $\lambda = 0.15$, $N_{\text{MC}} = 1500$, false-positive rate 0.05, seed 20260724. Arm B uses $d_{\text{cert}} = 64$, $n_{\text{cal}} = n_{\text{test}} = 24$, conjugate $z_1 = 0$, and perturbations fill $\pm 5\%$, pitch $\pm 3\%$, relay defocus 0–0.5 mm. All seven bars pass; the verdict is `TLSG_PASS` at commit `35d858e` (Table S2).

Bar 4 is the $1 : \sqrt{p} : p$ signature. Each bar-4 exponent is a single OLS log–log fit over $n = 4$ distinct $p \in \{10, 36, 136, 528\}$ (8 cells) at the frozen seed 20260724, with in-sample $R^2 = 0.003/0.983/0.9995$ (known/blind/estimation; deterministic recomputation from the committed `TLSG_partA.json` contours). A post-gate 10-seed bootstrap (`TLSG_BOOTSTRAP.json`; the `tlsg_A` runner unmodified, only the seed varied) attaches 95% confidence intervals and is reported as post-hoc robustness whatever it shows: matched exponent mean -0.011,

Table S2. TLSG gate: all seven frozen bars (`TLSG_RESULTS.json`, `@35d858e`). Values are quoted verbatim from the committed artifact.

#	Bar (frozen target)	Measured	Verdict
1	Matched-score 50%-power contours collapse vs $T\lambda$ (CV ≤ 0.20)	CV $= 0.1250$	pass
2	Blind contours collapse vs $T\lambda/\sqrt{p}$ (CV ≤ 0.25) AND fail vs $T\lambda$	$\mathrm{CV}_{\mathrm{scaled}} = 0.1242$; $\mathrm{CV}_{\mathrm{raw}} = 0.6283$	pass
3	Relative estimation risk collapses vs $T\lambda/p$ (CV ≤ 0.25)	CV $= 0.0348$ (mean risk $\cdot\, T\lambda/p = 0.985$)	pass
4	Fitted p-exponents (matched/blind/estimation): $0/\frac{1}{2}/1 \pm 0.15$	-0.004 / 0.444 / 1.010	pass
5	KT6 operating point below blind power 0.2 at FPR 0.05	blind power $=$ 0.0506 (10 $\times$ =0.0560, 100 $\times$ =0.1335; p=136, $T\lambda$=0.0866)	pass
6	Fresh calibration/test: $\geq$ 95% simultaneous coverage of $\sigma_d(\hat{L}) + \delta$	coverage $=$ 1.000 (σ_{64}=3.41 $\times$ 10^{-8}, δ=1.07 $\times$ 10^{-2}; worst fresh S_{64} leak 1.99 $\times$ 10^{-6})	pass
7	Conjugate plane: $\geq$ 64 code dims certified $< 10^{-4}$ on fresh law draws	fresh d@10^{-4} min $=$ 80 (mean 80.0; calibration d@10^{-4} $=$ d@10^{-5} $=$ 80)	pass

CI $[-0.044, 0.012]$ (contains 0); estimation mean 0.996, CI $[0.985, 1.008]$ (contains 1); blind mean 0.454, std 0.006, CI $[0.446, 0.465]$, which *excludes* the asymptotic $\frac{1}{2}$—a systematic finite-p deviation of $\approx$ 9% over the fit range rather than sampling scatter. The frozen gate verdict is unchanged: all 10 seeds pass every bar and lie inside the frozen ± 0.15 band, so the $0 : \frac{1}{2} : 1$ ladder-order separation is unambiguous even though the blind exponent sits systematically low. Bar 3's coefficient 0.985 is the Cramér–Rao floor p/T; bar 5 confirms the KT6 negative for the right reason (the operating point lies below the optimal blind tier). The oracle law itself is intact: the near-contact ratio $J_{\mathrm{cov}}/J_{\mathrm{mean}} \propto z_2^4$ has fitted exponents $[4.081, 4.054, 4.022, 3.989]$, but the blind median signal-to-floor ratio is $\approx 5.4 \times 10^{-3}$ (worst 5.25×10^{-3}), so the estimator is killed (KT6, `@635ff05`).

S5 Theorem-consistent honest negatives

The kills carry the same prominence as the positives. Two apparatus shortcuts were falsified under preregistered protocols and are recorded here in full.

Segmented-bucket negative (IT6, `@dade877`). The P^2 information law is real (fitted exponent 1.68–2.02; $P = 4$ buys 12–17$\times$), but segmenting the bucket *breaks the mean wall*: the per-segment discriminability d' grows from 0.8 to 1.8 at $z_2 = 5$, so beyond-band change floods the segmented mean channel. The single, whole-aperture bucket stands; segmentation is not a valid capacity multiplier. Verdict: KILL.

Coincidence-veto negative (IT7, `@e1546ac`). A contact-arm coincidence veto is a valid medium-event discriminator—the contact arm is exactly medium-blind (AUC $=$ 0.5009) and the medium false-alarm rate is repaired 17.8$\times$ (from 0.9965 to 0.056)—but it is an invalid beyond-band repair, retaining only 24% of scene power, because the same wall that supplies specificity starves the contact arm of beyond-band signal. It belongs in the bench manual as a theorem-consistent negative control, not as a method. Verdict: KILL. This is why no strong-specificity wording is claimed, and the semiparametric projection remains a quarantined future method until estimator-layer certification is solved.

Future objects (deferred, not claimed). A task-dependent étendue law $N_* = [\nu/(2-\nu)]\, n$ is only partially collapsed (KT5, `@752114a`: N_{eff} grows $46 \to 811$, normalized CV 0.64), and coherence-rank unsaturation persists with a thin screen holding $R \approx 110$ without an $R \to 1$ collapse (KT4, `@494b46b`). Both are gated future work (named gates étendue-matching and COHERENCE-CROSSOVER); neither supports a claim here. The memory-effect phase diagram requires a genuine thickness/stacking axis and is deferred.

Table S3. Frozen-source provenance (commit at freeze). Full field-level map in `paper2_optica/CLAIM_SOURCE_MATRIX.md`.

Artifact	Commit	Role
TLSG_RESULTS.json	`35d858e`	three-ledger scaling + calibration transfer (gate)
TLSG_BOOTSTRAP.json	`3646754`	post-gate 10-seed exponent CIs (post-hoc robustness)
KT1 shell sweep	`19338a4`	support/field wall 6.6×10^{-16} + breakers
IT1 two-wall jet	`f13a323`	energy/unitary wall 2.18×10^{-16}; NA-clip ε^1
KT2 jet transmutation	`05f1029`	projected slope 4.00; jet flow
KT3 orthogonalized sentinel	`2430166`	code-space projection, 86–99% retained
IT4 / IT4b	`33b8ef2` / `5666d1e`	capacity spectrum (80/35/24 dims)
IT5 DPSS concentration	`c8a0487`	DPSS route 2.4×10^{-4}; $80 \to 24$
KT6 channel-ratio ranging	`635ff05`	third-ledger boundary (oracle z_2^4)
IT2 / IT3 / IT8	`acfcda0` / `6fb5149` / `6fb5149`	floor decomposition, leak law, PWM
IT6 / IT7	`dade877` / `e1546ac`	honest negatives (Section S5)
WAVE_TWIN	`0c3b0d9`	COMSOL 15.5%, $M^{1.8}$, matched cell $768 \in [513, 2996]$
JET_TEST / SCRAMBLE_EXT	`1bf29f1` / `ed7a1e0`	companion jet/scrambling anchors

S6 Model and protocol capsule

Engines and shared configuration. The wall/jet exams use the exact Kronecker speckle generator (fully developed circular-complex-Gaussian field, factorized grain $\otimes$ code covariance), grid 2048 for IT1/KT1 in `complex128`/ `float64`, with code counts $M \in \{8, 12, 16, 32\}$ and 200–700 banks per test. The blindness-capacity and route tests use the SVD/DPSS leak operators at the relay-conjugate plane. The COMSOL full-Maxwell twin (Wave Optics, 2D TE, COMSOL 6.3; a COMSOL license is required to reproduce it) *scopes* the thin-phase screen rather than validating it wholesale: over $n = 8$ realizations the ensemble speckle contrast agrees to 15.5% (0.467 versus 0.395) while the speckle grain size disagrees by 65% (3.88 μm versus 1.35 μm, a 2.9$\times$ overestimate). The per-realization contrast disagreements $|\Delta|$/FEM span 11.6%–60.9% (median $\approx$ 19%); the reduced model transfers contrast statistics but not grain scale (WAVE_TWIN, `@0c3b0d9`; per-realization values in `COMSOL_developed_results.json`). No reconstruction is performed anywhere.

Scaling gate. Arm A and arm B parameters are in Section S4.3; the single seed is 20260724; there was one sealed run and no repair round.

Provenance. Every number in the article and this Supplement traces to one committed frozen artifact (Table S3); the extracted verbatim copies read by the figure scripts live in `figures/_frozen_sources/`. The working tree is clean for all cited artifact directories.

Data availability. The committed artifacts, claim–source matrix, and figure scripts are available at [URL].

References

[1] [AUTHORS], Strong optical disorder erases images but preserves local change observability, companion Letter (2026).